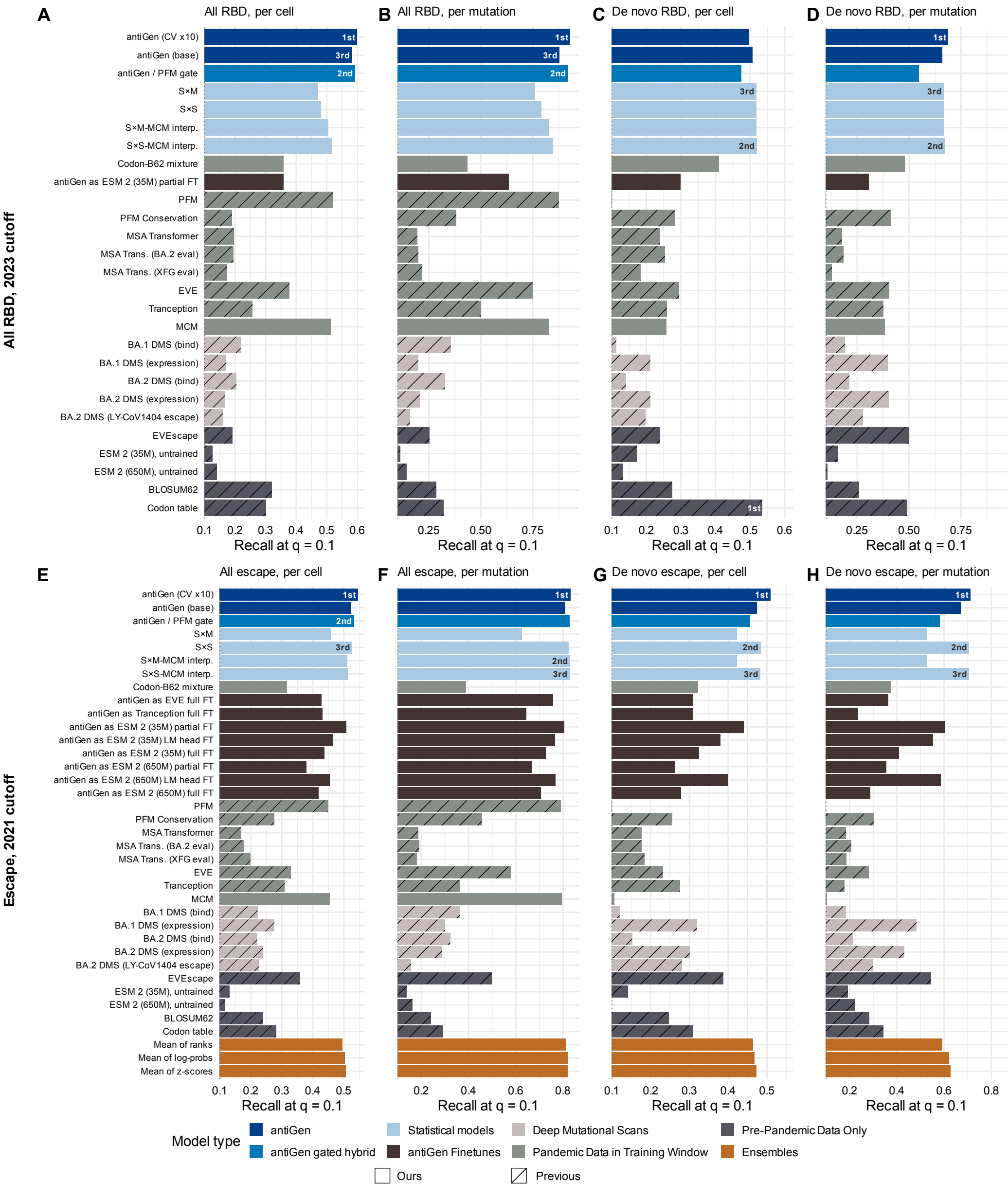

A
All RBD, per cell
B
All RBD, per mutation
C
De novo RBD, per cell
D
De novo RBD, per mutation
All RBD, 2023 cutoff
antiGen (CV x10)
antiGen (base)
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
Codon-B62 mixture
antiGen as ESM 2 (35M) partial FT
PFM
PFM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
MCM
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY-CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1 0.2 0.3 0.4 0.5 0.6
0.25 0.50 0.75
Recall at q = 0.1
E
All escape, per cell
F
All escape, per mutation
G
De novo escape, per cell
H
De novo escape, per mutation
Escape, 2021 cutoff
antiGen (CV x10)
antiGen (base)
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
Codon-B62 mixture
antiGen as EVE full FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) full FT
PFM
PFM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
MCM
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY-CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
Mean of ranks
Mean of log-probs
Mean of z-scores
1st
2nd
3rd
0.1 0.2 0.3 0.4 0.5
0.2 0.4 0.6 0.8
Recall at q = 0.1
Model type
antiGen
antiGen gated hybrid
Statistical models
antiGen Finetunes
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ensembles
Ours
Previous